



CL 299 presentation 8

LNP Manufacturing and Process Simulation

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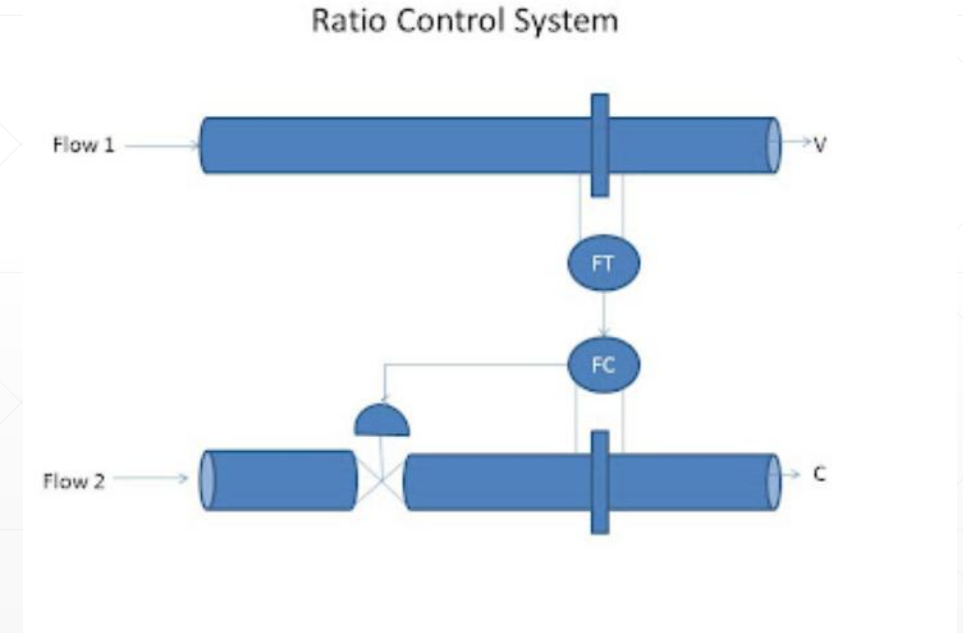
TYPES OF CONTROLLER USED IN P&ID

RATIO CONTROLLER

- A specialized control strategy used to maintain a fixed ratio between two or more process streams.
- Select a **Master Stream**.
- Divide the flow according to Master stream.
- The controller calculates the setpoint for the follower stream.

$$F_{follower} = Ratio * F_{Master}$$

- Useful → Wrong ratio is major risk.



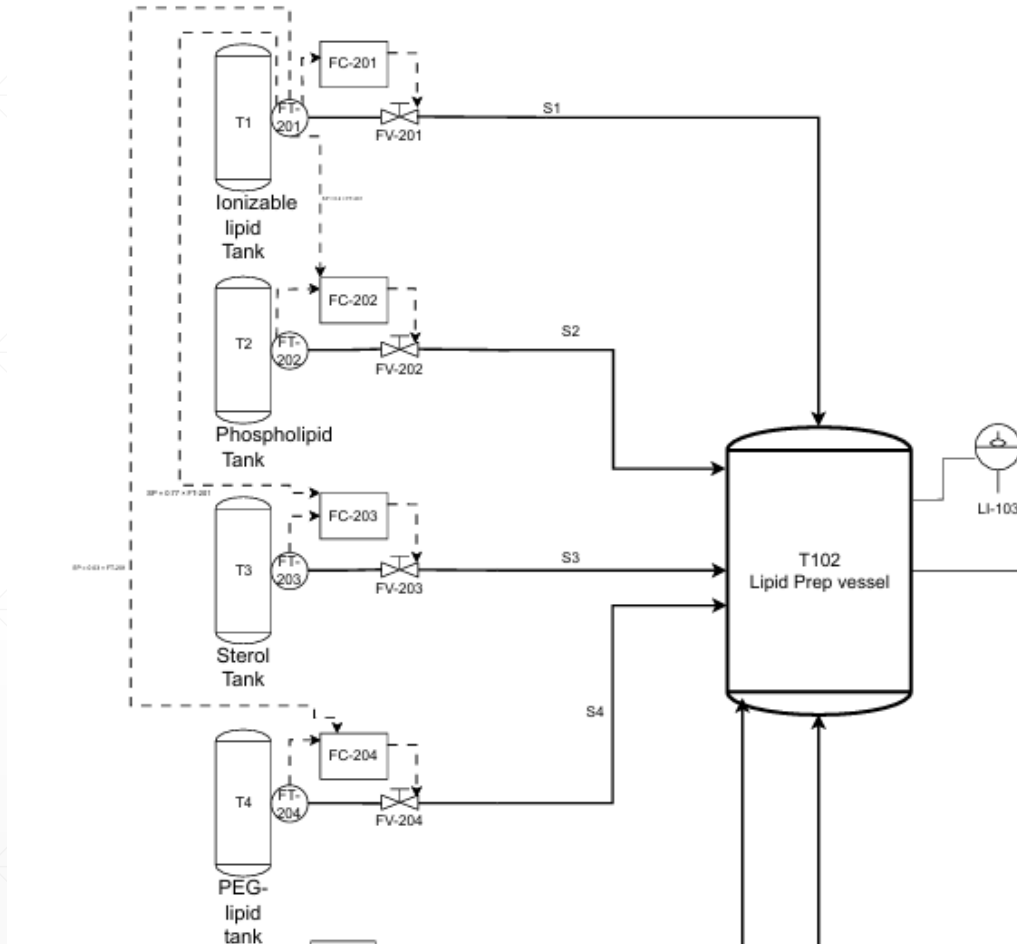
HOW WE USE RATIO CONTROLLER ?

- S1 as master stream → Measure.
- Measured stream to be adjusted.
- Send signals to other controllers.
- Then divide the flow.

$$S2 = 0.2 * S1$$

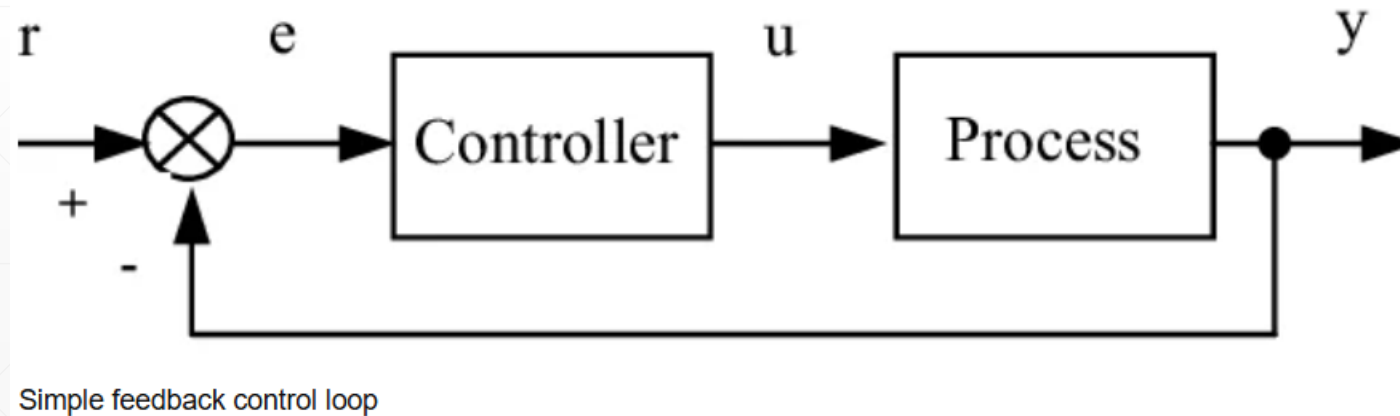
$$S3 = 0.77 * S1$$

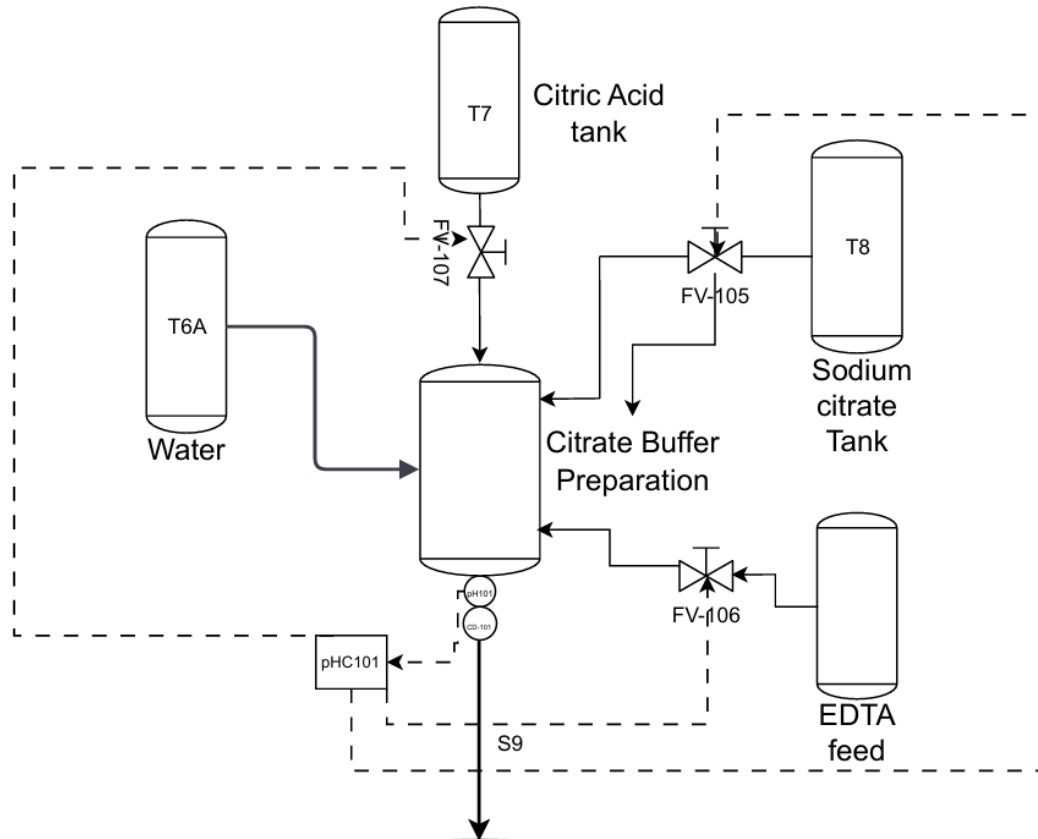
$$S4 = 0.03 * S1$$



Feedback Controller

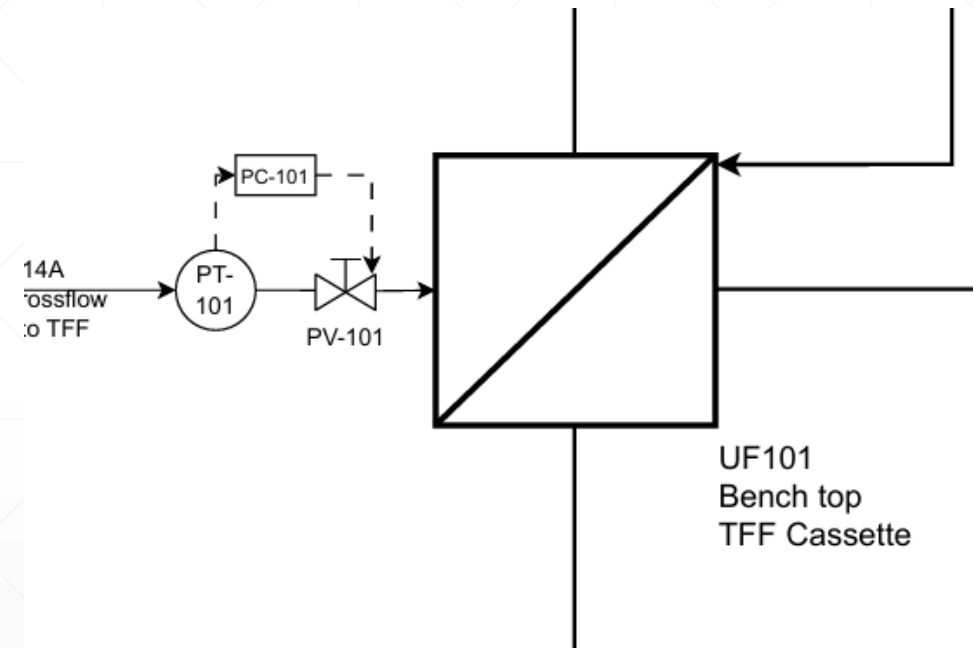
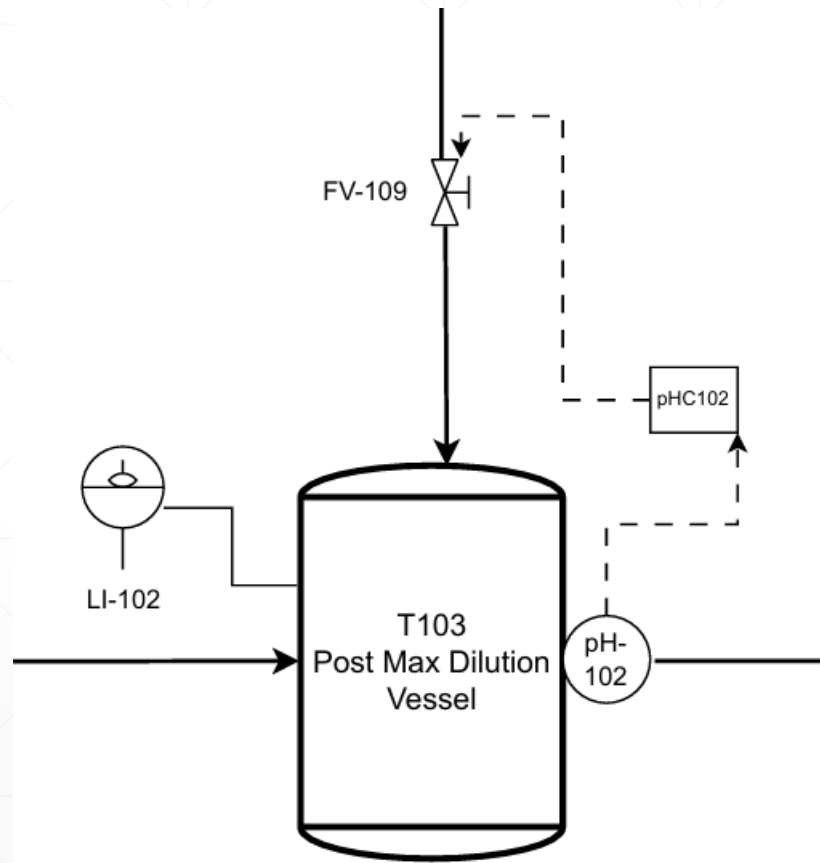
- Closed-loop control strategy.
- output of a process is continuously measured by a sensor, compared to a desired setpoint.
- the error is used to automatically adjust the manipulated variable to bring the process back to the setpoint.





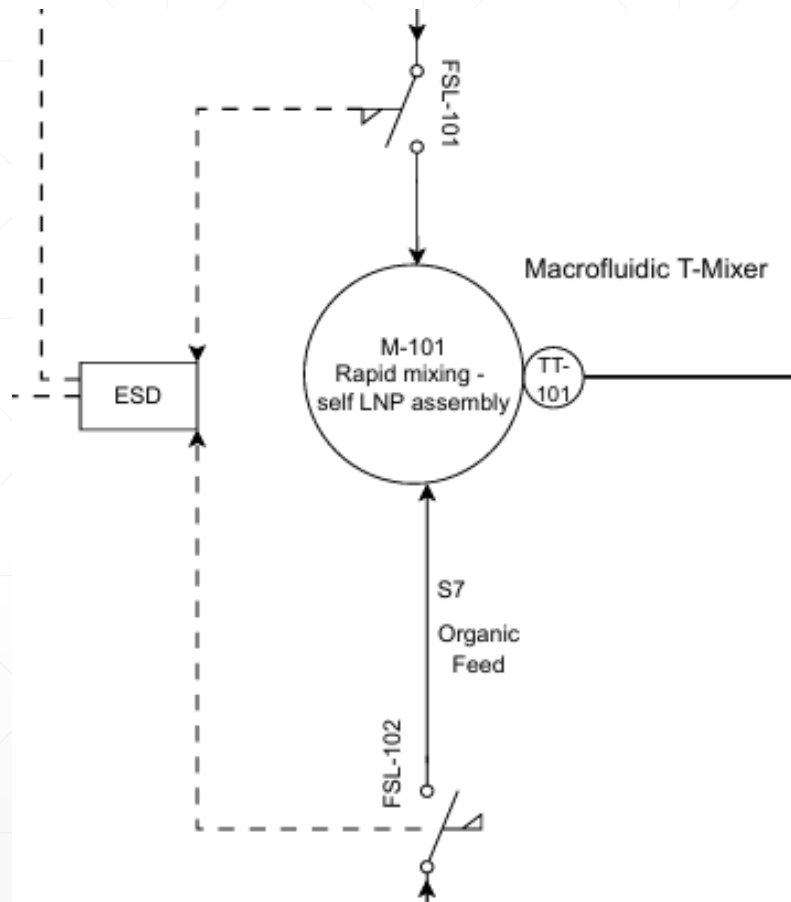
HOW WE USE FEEDBACK CONTROLLER?

- pH transmitter continuously measures solution Ph
- Controller compares measured pH to setpoint and generates an error signal
- Error drives acid or base valve to bring pH back to setpoint



On/Off (bang-bang) Controller

- Close loop control strategy.
- The controller output has only two possible states: Fully ON or Fully OFF
- A switching action is triggered when the measured variable crosses a preset threshold.
- Implemented as part of a Safety Instrumented System (SIS).



HOW WE USE On/Off CONTROLLER?

- FSL-101 or FSL-102 drops below trip setpoint.
- ESD activates.
- full process shutdown

Resources

- G. Dragffy, “The design of a highly reliable safety critical emergency shutdown (ESD) system,” *Reliability Engineering & System Safety*, 1998.
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THANK YOU!!!